NATIONAL MATHEMATICS CONTEST
O-LEVEL PAPER 1 QUESTIONS AND SOLUTIONS.

SATURDAY 5TH APRIL, 2025 9:00 AM - 12:15 PM

SECTION A: 5 marks each

1. Let S_n be the sum of the first n terms of the sequence (a_n). If $S_n = 2^n$ for all $n \geq 1$, what is the value of a_{2025}?

   Solution

   By using the general formula $S_n = 2^n$ for all $n \geq 1$, we find that

   $$a_{2025} = 2^{2025} - 2^{2024} = 2^{2024}(2-1) = 2^{2024}.$$

2. Amara has 13 blue pens, 5 green pens and 9 red pens in her bag. She draws one pen at a time and puts it aside. How many times should she draw to ensure that she has at least two pens of each colour?

   Solution

   The worst case scenario is that Amara first draws all the 13 blue pens followed by 9 red pens and hence still does not have two of each colour. Drawing 23rd and 24th pen forces at least two pens of each colour.

3. Work out

   $$1000 - 999 + 998 - 997 + \ldots + 4 - 3 + 2 - 1.$$

   Solution

   It is clear that if we pair the numbers from the start, the sum in each pair, or rather the difference in each pair, is 1. Hence, the sum is 500 as there are 500 pairs which have a difference of 1.

4. A number is called Beprisque if it is the only natural number (positive whole number) between a prime number and a perfect square (e.g. 8 and 10 are Beprisque but 12 is not). Find all the two-digit Beprisque numbers ?

   Solution

   Since there are less square numbers than primes below 100, we just test each square number. The perfect squares are $1, 4, 9, 16, 25, 36, 49, 64, 81, 100$ and we need to find how many of these are such that the number 2 away from them is a prime number. None of the even perfect squares can work as the number 2 away from them cannot be prime. Hence we are left with $9, 25, 49, 81$. (Note that 1 and 4 are left out since 2 numbers away from tem do not form a two digit number). The numbers $11, 23, 47, 79$ and 83 are all prime and consequently the two-digit Beprisque numbers are $10, 24, 48, 80$ and 82

5. Solve for x in the equation

   $$\sqrt{x\sqrt{x\sqrt{x\sqrt{x\sqrt{x}}}}} = 12.$$

   Solution 1

   - By squaring continuously both sides until all the radicals are removed we get:

     $$x^{16} \cdot x^8 \cdot x^4 \cdot x^2 \cdot x^1 = ((((12^2)^2)^2)^2)^2 \quad \Rightarrow x^{31} = 12^{32}.$$

   - Thus, applying the simple rules of algebra, we get that $\boxed{x = 12^{\frac{32}{31}}}$.

   Solution 2

   - The equation can be rewritten as:

     $$x^{\frac{1}{2}}(x^{\frac{1}{2}})^{\frac{1}{2}}((x^{\frac{1}{2}})^{\frac{1}{2}})^{\frac{1}{2}}(((x^{\frac{1}{2}})^{\frac{1}{2}})^{\frac{1}{2}})^{\frac{1}{2}}((((x^{\frac{1}{2}})^{\frac{1}{2}})^{\frac{1}{2}})^{\frac{1}{2}})^{\frac{1}{2}} = 12 \quad =$$

   - Thus, applying simple algebra rules: $x^{\frac{31}{32}} = 12$ or $\boxed{x = 12^{\frac{32}{31}}}$.

6. Find the sum of the digits of the integer equal to

   $$777777777777777^2 - 222222222222223^2.$$

   Solution

   Using the property of difference of squares, we have $x^2 - y^2 = (x-y)(x+y)$. Now, $777777777777777^2 - 222222222222223^2$=(777 777 777 777 777- 222 222 222 222 223)(777 777 777 777 777+ 222 222 222 222 223) = 555 555 555 555 554 $\times$1000000000000000 = 555555555555554000000000000000 and hence the sum of the digits is $15 \times 5 - 1 = 74$.

7. Teddy mistakenly wrote $2^m \cdot \sqrt{\frac{1}{4096}}$ as $2 \cdot \sqrt[m]{\frac{1}{4096}}$. Find the value of m for which these two expressions have the same value?

   Solution

   Notice that $4096 = 2^{12}$, so the first expression is $2^m(2^{-12})^{1/2} = 2^{m-6}$, and the second expression is $2(2^{-12})^{1/m} = 2^{1-\frac{12}{m}}$. These two expressions have the same value if and only if $m - 6 = 1 - \frac{12}{m}$. Multiplying both sides of this equation by m, rearranging terms and factoring gives $(m-4)(m-4) = 0$. The solutions of this equation are $m = 3$ and $m = 4$. The requested sum of possible values for m is $3 + 4 = 7$.

8. Linda leaves her house for work at exactly 8 : 00A.M every morning. When she averages 40km/hr, she arrives at her workplace three minutes late. When she averages 60 Km/hr, she arrives three minutes early. At what average speed, in kilometres per hour, should Linda drive to arrive at her workplace precisely on time?

   Solution

   Let t be the number of hours Linda must travel to arrive on time. Since three minutes is the same as 0.05 hours, we have

   $$40(t + 0.05) = 60(t - 0.05).$$

   Thus, $40t + 2 = 60t - 3$, so that $t = 0.25$.

   The distance from her home to work is $40(0.25 + 0.05) = 12$km. Therefore, her average speed should be $\frac{12}{0.25} = 48$Km/hr.

   Alternatively

   Let d be the distance from Linda's house to work, and let s be the desired average speed. Then the desired driving time is d/s. Since $d/60$ is three minutes too short and $d/40$ is three minutes too long, the desired time must be the average, so

   $$\frac{d}{s} = \frac{1}{2}\left(\frac{d}{60} + \frac{d}{40}\right).$$

   This implies that $s = 48$km/hr.

9. Jokers always lie. Clowns always tell the truth. A group of four, each of whom is a Joker or a Clown, make the following statements about each other:

   P says, "Q always lies";

   Q says, "R always lies";

   R says, "P always tells the truth";

   S says, "Exactly two of P, Q and R are Jokers".

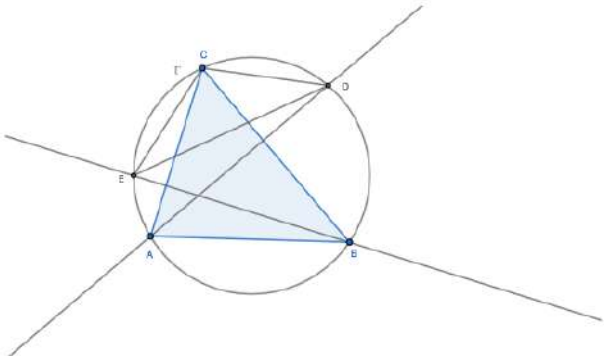
   How many of P, Q, R and S are Clowns?

   Solution

   Suppose P is a Clown and hence is telling the truth. So Q is lying and hence is a Joker, R is telling the truth and so is a Clown, and S is telling a lie and hence is a Joker. Therefore, in this case, there are two Clowns. Now suppose P is a Joker and hence is telling a lie. Then Q always tells the truth and hence is a Clown, R is telling a lie and hence is a Joker, and S is telling the truth and hence is a Clown. Therefore, in this case, there are two Clowns. It follows that in any case there are two Clowns.

10. Let ABC be an acute angled triangle with circumcircle Γ. The perpendicular from A to BC intersects Γ at D, and the perpendicular from B to AC intersects Γ at E. Prove that if $\overline{AB} = \overline{DE}$, then $\angle ACB = 60^o$.

    Solution Consider the diagram below for illustration of the problem:



- Since AB and ED are equal chords in the same circle, we either have $\angle ACB = \angle ECD$ or $\angle ACB + \angle ECD = 180^o$.
- We compute $\angle ECD$ as follows: $\angle ECD = \angle ECA + \angle ACB + \angle BCD = \angle EBA + \angle ACB + \angle BAD$

$=90^o - \angle BAC + \angle ACB + 90^o - \angle ABC$
$=(180^o - \angle BAC - \angle ABC) +$
$\angle ACB = 2\angle ACB.$

- If $\angle ACB = \angle ECD$, then $\angle ACB = 0^o$ an obvious contradiction. Thus, $\angle ACB + \angle ECD = 180^o$, that is, $3\angle ACB = 180^o$, whence $\boxed{\angle ACB = 60^o}$.

SECTION B: 10 marks each

11. Granny gave away her entire collection of spoons to three of her grandchildren Hellen, Jayden and Samuel. Hellen received 8 more than a third of the total; Jayden received 8 more than a third of what was then left; finally Samuel received 8 more than a third of what was then left. What is the total number of spoons which were in Granny's collection?

    Solution

    Suppose Granny began with x spoons. She gives $\frac{1}{3}x + 8$ spoons to Hellen. Therefore the number of spoons she has left is

    $$x - \left(\frac{1}{3}x + 8\right) = \frac{2}{8}x - 8.$$

    The same formula applies at the second stage, with x replaced by $\frac{2}{8}x - 8$. So, after giving spoons to Jayden, Granny is left with

    $$\frac{2}{8}\left(\frac{2}{8}x - 8\right) - 8.$$

Similarly, the same formula applies at the third stage with x replaced by $\frac{2}{8}\left(\frac{2}{8}x-8\right)-8$. So, after giving spoons to Samuel, Granny is left with

$$\frac{2}{3}\left(\frac{2}{8}\left(\frac{2}{8}x-8\right)-8\right)-8.$$

Now

$$\frac{2}{3}\left(\frac{2}{8}\left(\frac{2}{8}x-8\right)-8\right)-8=\frac{8}{27}x-\frac{8\times 19}{9}.$$

Therefore, Granny gives away her entire collection of spoons provided that

$$\frac{8}{27}x-\frac{8\times 19}{9}=0$$

This gives

$$\frac{8}{27}x=\frac{8\times 19}{9}$$

so that

$$x=\frac{27}{8}\times\frac{8\times 19}{9}=57.$$

Therefore Granny began with 57 spoons.

ALTERNATIVELY

In this method we simplify the algebra by working backwards. We have seen in the previous Method that if Granny begins with x spoons, after giving away 8 more than a third of her spoons, she is left with $\frac{2}{3}x-8$ spoons. We put $y=\frac{2}{3}x-8$. This gives $\frac{3}{2}(y+8)$. Therefore if Granny is left with y spoons, she began with

$$\frac{3}{2}(y+8) \qquad (1.1)$$

spoons.

Hence by putting $y=0$ in the formula (1), we see that Granny had 3 $2(0+8)=12$ spoons before giving all her remaining spoons to Samuel.

Similarly, by putting $y=12$ in (1), we see that Granny had $\frac{3}{2}(12+8)=30$ spoons before giving spoons to Jayden. Finally, putting y = 30 in (1), we see that before giving spoons to Hellen, Granny had $\frac{3}{2}(30+8)=57$ spoons.

12. Let n be a positive integer. Find all possible values of n in the equation:

$$3^n-2^n=65.$$

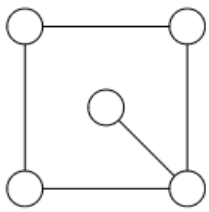
Solution

- The equation can be rewritten as $\left(3^{\frac{n}{2}}\right)^2-\left(2^{\frac{n}{2}}\right)^2=65$ or $x^2-y^2=65$, where $x=3^{\frac{n}{2}}>0$ and $y=2^{\frac{n}{2}}>0$.
- Now, taking the difference of two squares and factoring the RHS of the resulting equations yields: $(x-y)(x+y)=65=5\cdot 13=1\cdot 65$.
- Since both x and y are positive, we have the following pairs of equations:

$$x-y=5,\quad x+y=13; \qquad (1.2)$$
$$x-y=1,\quad x+y=65. \qquad (1.3)$$

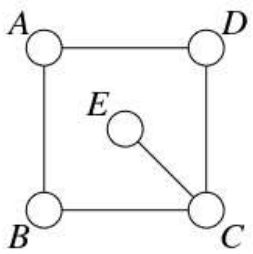
- Solving the equations in (1.2) simultaneously, yields: $(x,y)=(9,4)$ and those in (1.3) yiled: $(x,y)=(33,32)$.
- Thus, $3^{\frac{n}{2}}=9=3^2$ or $\frac{n}{2}=2$, that is, $n=4$ and $2^{\frac{n}{2}}=4=2^2$ or $\frac{n}{2}=2$, that is, $n=4$. On checking with the original equation, this value works.
- Also, $3^{\frac{n}{2}}=33$ or $n=2\frac{\log 33}{\log 3}\notin\mathbb{Z}$ (not an integer), and $2^{\frac{n}{2}}=32=2^5$, so $\frac{n}{2}=5$, which implies that $n=10$. On checking with the orignal equation, this value fails to work since $3^{10}-2^{10}=58025\neq 65$. Thus, we have no other solutions apart from $\boxed{n=4}$.

13. The figure on the right shows five circles connected by five line segments. Three colours are available to colour these circles. In how many different ways is it possible to colour all five circles so that circles which are connected by a line segment are coloured differently?



Solution

- We first look at the number of different ways of colouring the four circles at the corners of the square.
- If circles B and D are given the same colour, then they may be coloured in three different ways. For each of these, two colours are available for both circle A and circle C.
- Therefore, if B and D have the same colour, then the four corner squares may be coloured in $3\cdot 2\cdot 2=12$ different ways.



- If circles B and D are coloured differently, then we can choose three colours for B and, for each of these, two different colours are available for D. So B and D may then be coloured in $2 \cdot 3 = 6$ different ways.
- Now, however, only one colour is available for circles A and C, so if circles B and D are coloured differently then there are six ways of colouring the corner circles.
- Hence in total the four corner circles may be coloured in $12 + 6 = 18$ different ways. For each of these 18 possibilities, circle E must be coloured differently from circle C and thus two colours are available for it.
- Therefore, in total, there are $2 \cdot 18 = 36$ different ways of colouring the five circles so that connected circles are coloured differently.

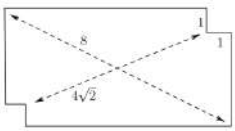
14. Rovers, United, City and Wanderers played against each other once in a Hockey tournament. The results are shown in the Table on the right. What was the score in the match between Rovers and United?

Team	Win	Draw	Loss	Goals for	Goals against
Rovers	3	0	0	5	0
United	0	2	1	3	6
City	1	1	1	4	4
Wanderers	0	1	2	0	2

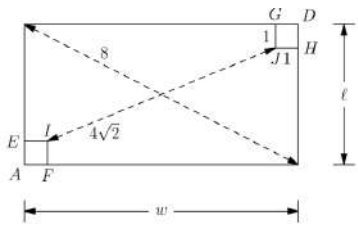
Solution

Note first that United drew two of their matches and lost one match. Also they scored three fewer goals than were scored against them. In the matches they drew they will have scored the same number of goals as their opponents. Hence, in the match which they lost, United lost by a margin of three goals. Rovers won all their matches. Hence Rovers beat United. Also, Rovers had no goals scored against them. Therefore United lost to Rovers by a margin of three goals and without scoring a goal. It follows that the score in the match between Rovers and United was a $3 - 0$ win for Rovers.

15. Dan finds a rectangular index card and measures its diagonal to be 8 centimeters. He then cuts out equal squares of side 1 cm at two opposite corners of the index card and measures the distance between the two closest vertices of these squares to be $4\sqrt{2}$ cm, as shown on the right. What is the area of the original index card?



Solution



Label the bottom left corner of the larger rectangle (without the square cut out) as A and the top right as D. w is the width of the rectangle and ℓ is the length. Now we have vertices E, F, G, H as vertices of the irregular octagon created by cutting out the squares. Let I, J be the two closest vertices formed by the squares. The distance between the two closest vertices of the squares is thus $IJ = \left(4\sqrt{2}\right)$. Substituting, we get

$$(IJ)^2 = (w-2)^2 + (\ell - 2)^2 = \left(4\sqrt{2}\right)^2 = 32$$

$\implies w^2 + \ell^2 - 4w - 4\ell = 24$.Using the fact that the diagonal of the rectangle is 8, we get

$$w^2 + \ell^2 = 64.$$

Subtracting the first equation from the second equation, we get

$$4w + 4\ell = 40 \implies w + \ell = 10.$$

Squaring yields $w^2 + 2w\ell + \ell^2 = 100$. Subtracting the second equation from this, we get $2w\ell = 36$, and thus area of the original rectangle is $w\ell = 18$.

END